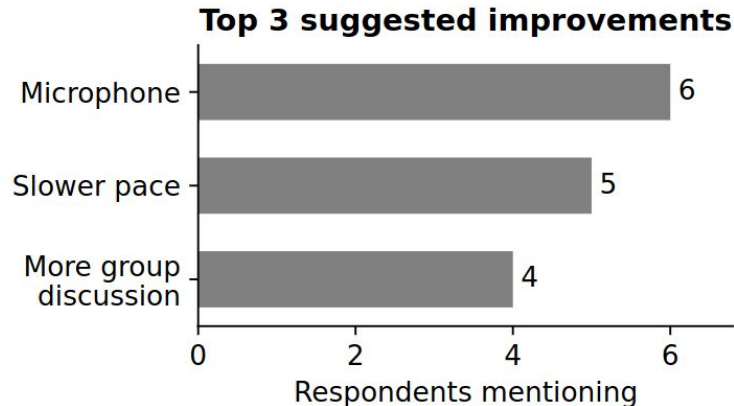
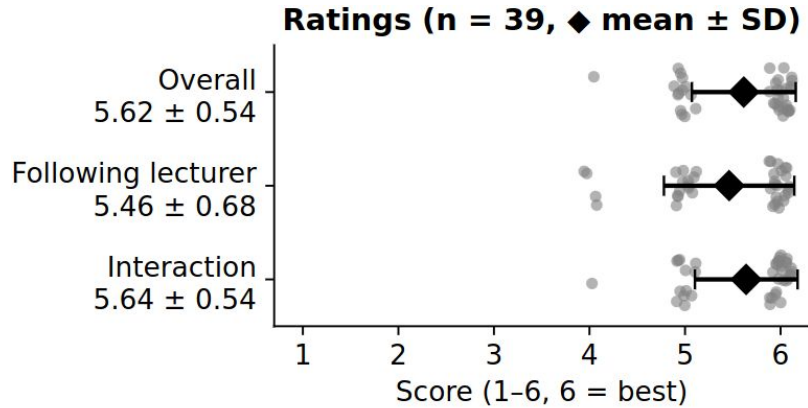


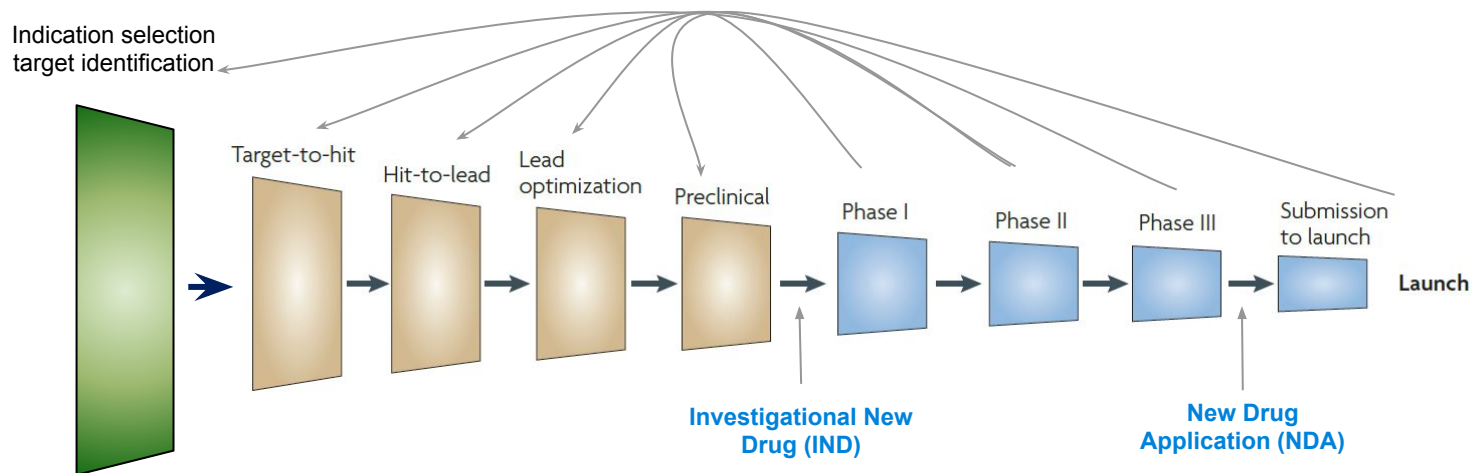
Feedback and recurring questions



- *If the patients stop taking the drug, will they regain the weight? Do patients have to keep taking it forever? Is it a kind of dependency - which, on the other side, would make it a 'good' business model?*
- *Why does semaglutide lead to more weight loss than liraglutide, even at a lower dose (2.4 mg vs 3 mg)? What happens to liraglutide? Is it obsolete? What are the differences in the design of the trials?*
- *What is the actual main goal of drug research: profit, or helping people in need? Would companies keep a drug off the market if it would hurt sales of their other drugs? How do companies make these expensive medicines cheaper for everyone?*

AMIDD 2026 Lecture 2:

The *What*, *Who*, and *How* of drug discovery



Dr. Jitao David Zhang, Computational Biologist

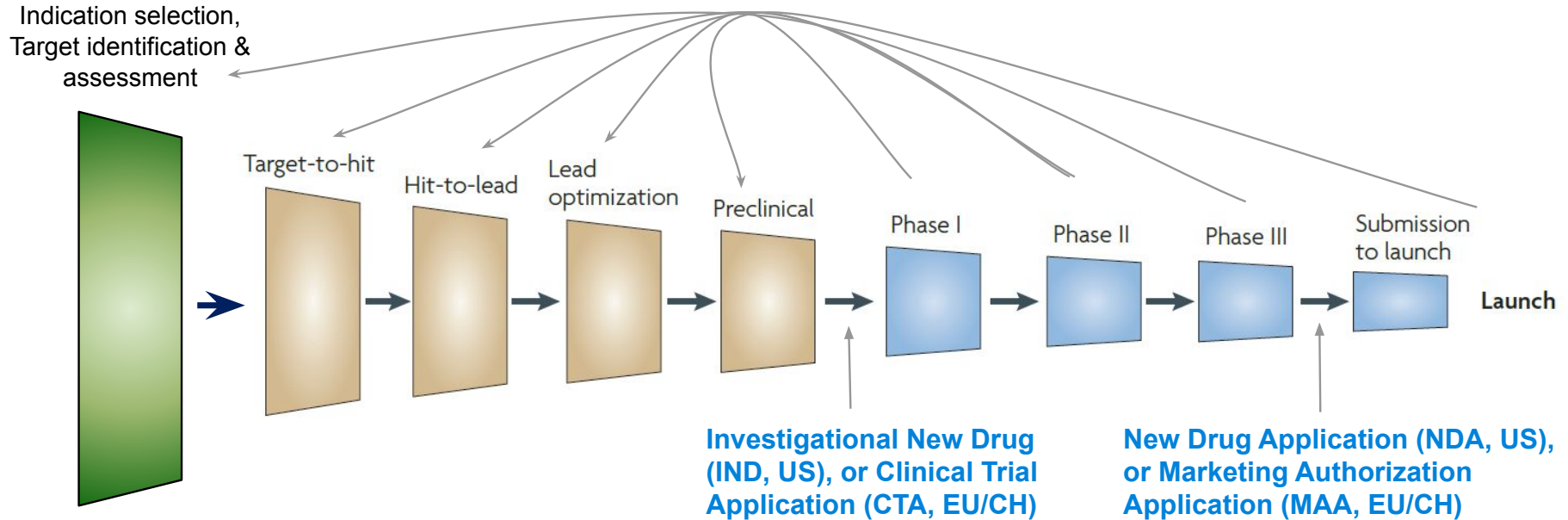
¹ Computational Sciences Center of Excellence (CS CoE), Roche Innovation Center Basel, F. Hoffmann-La Roche;

² Department of Mathematics and Computer Sciences, University of Basel

Outline

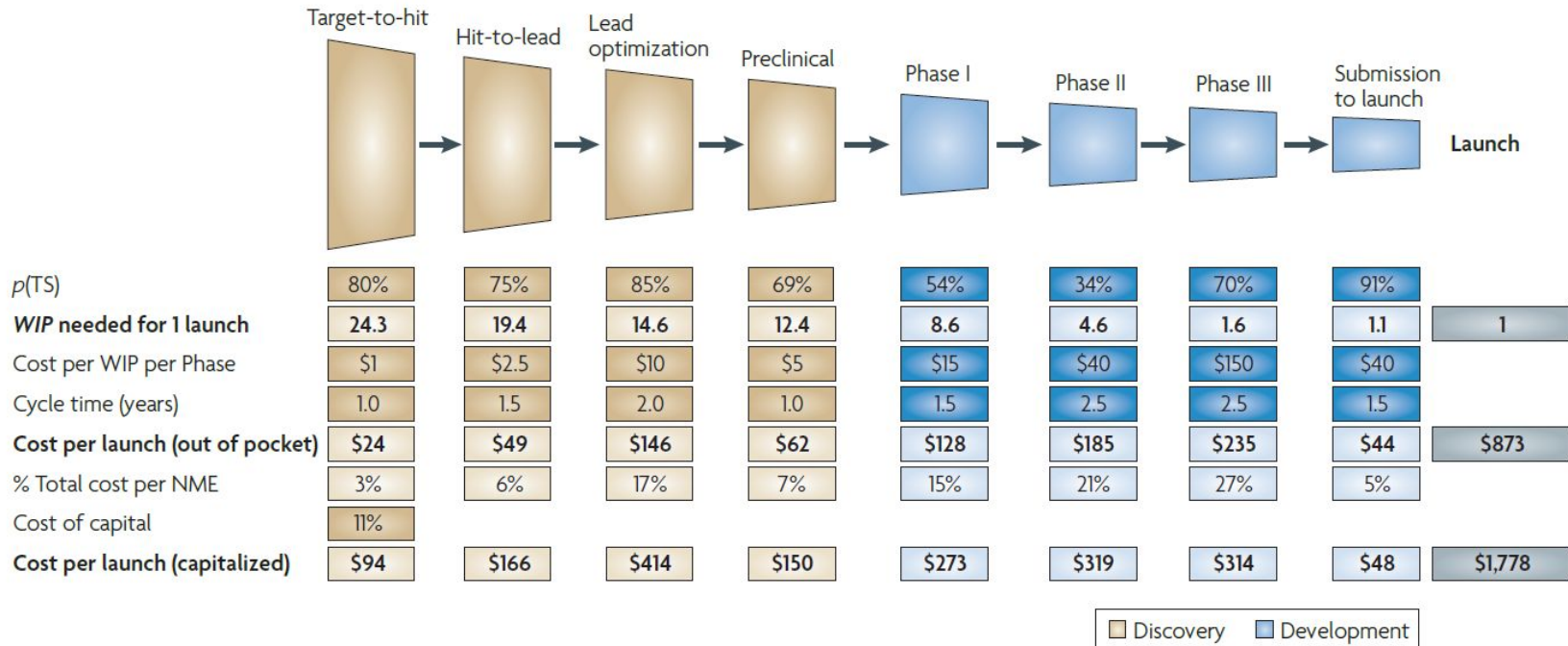
1. *How*: Linear and multi-scale views of drug discovery
2. *What*: Modalities
3. *Who*: A role game

The linear view of drug discovery



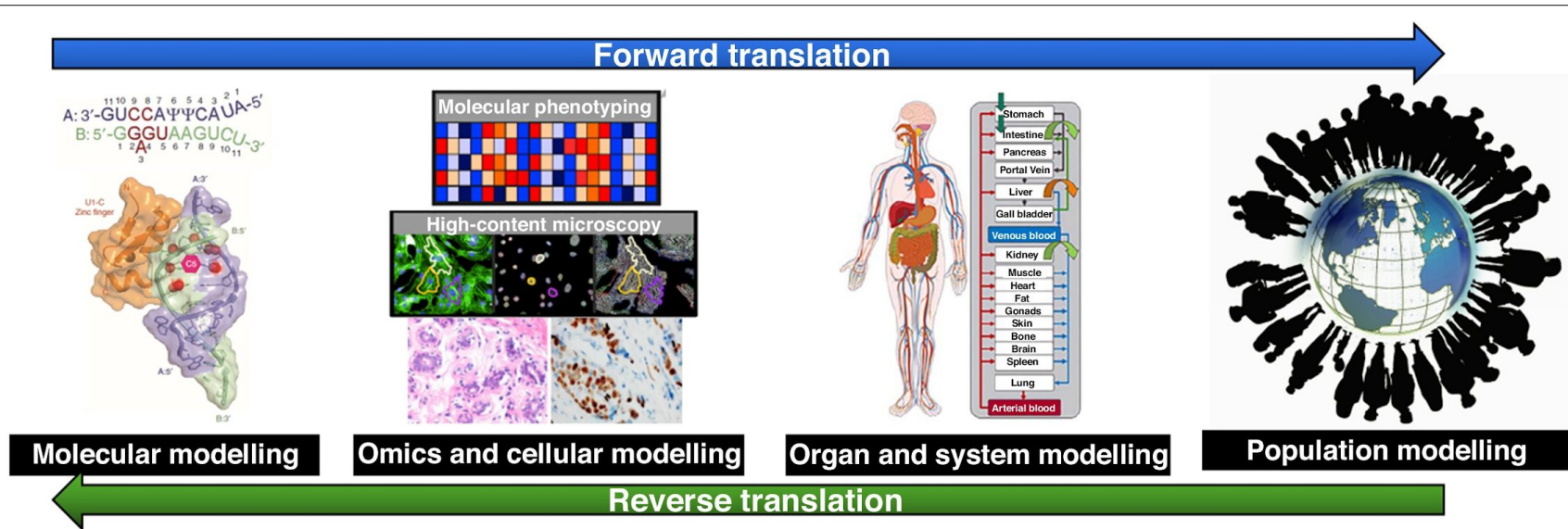
Adapted from Paul *et al.* "How to Improve R&D Productivity: The Pharmaceutical Industry's Grand Challenge." *Nature Reviews Drug Discovery*, 2010

Risks and costs associated with each stage of the linear view of drug discovery



$p(TS)$: probability of technical success. **WIP**: work in progress; **Capitalized cost**: Out-of-pocket cost corrected for cost of capital, standard for long-term investments; **Out-of-pocket cost**: total cost required to expect one drug launch, taking into account attrition, but not the cost of capital; **Cost of capital**: annual rate of return expected by investors based on the level of risk of the investment. Paul *et al.*, Nature Reviews Drug Discovery, 2010.

The multiscale modelling view of drug discovery



Drug Discovery Today

Zhang, Jitao David, Lisa Sach-Peltason, Christian Kramer, Ken Wang, and Martin Ebeling. 2020. "Multiscale Modelling of Drug Mechanism and Safety." *Drug Discovery Today* 25 (3): 519–34. <https://doi.org/10.1016/j.drudis.2019.12.009>.

Glucagon-like peptide 1 (GLP-1) and chemical derivatives

GLP-1 is a 30- or 31-amino-acid peptide. It is derived from a protein (preproglucagon) encoded by the gene GCG in the human genome.

Endogenous GLP-1 has a half-life of about 2 minutes, making the natural form a poor drug candidate due to its pharmacokinetic (PK) behaviour.

Synthesized agonists have similar sequences but much longer half-lives due to chemical modifications.

Examples include:

- Semaglutide (top, Novo Nordisk, approved, half-life: about 1 week)
- Taspoglutide (bottom panel, tested in Phase III trials, discontinuation due to hypersensitivity and poor GI tolerability)

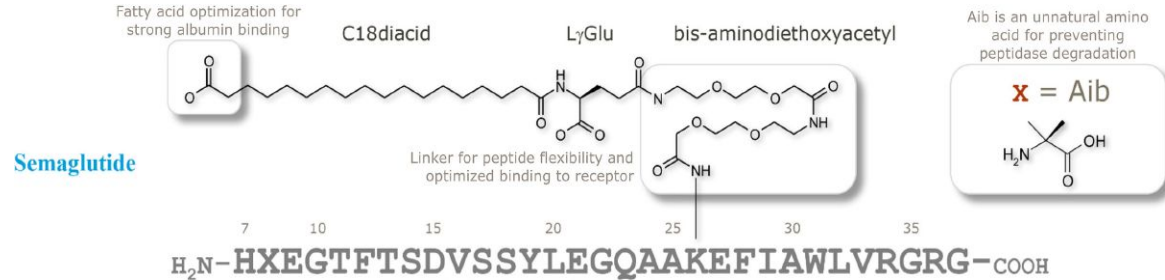


Figure from [Knudsen et al., 2019](#)

SEMAGLUTIDE	1 HXEGTFTSDVSSYLEGQAAKEFIAWLVRGRG	31
GLP-1	1 HAEGTFTSDVSSYLEGQAAKEFIAWLVRGRG	31
TASPOGLUTIDE	1 HXEGTFTSDVSSYLEGQAAKEFIAWLVRGR	30
GLP-1	1 HAEGTFTSDVSSYLEGQAAKEFIAWLVRGR	30

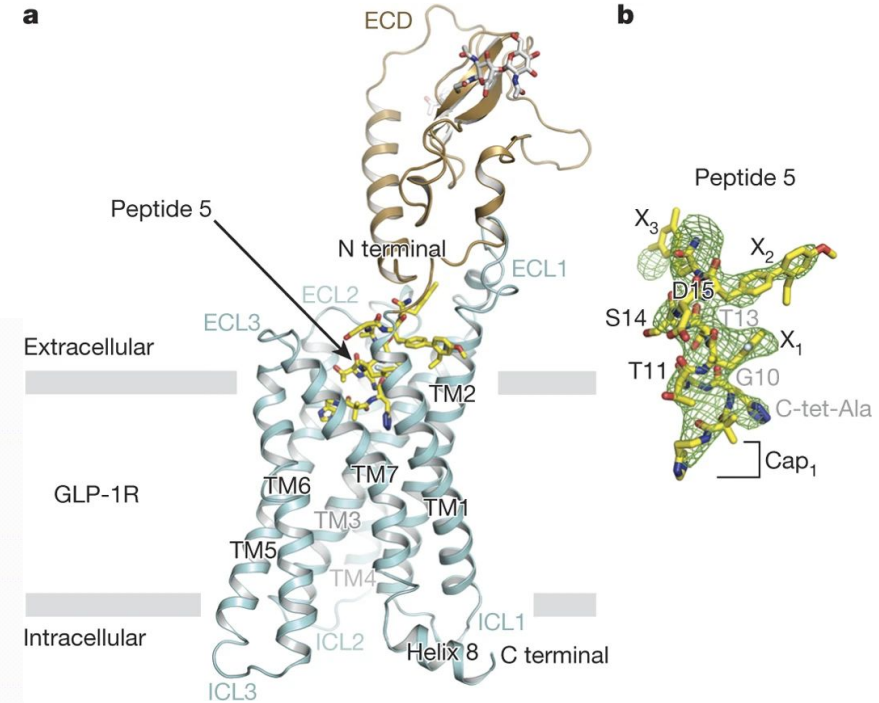
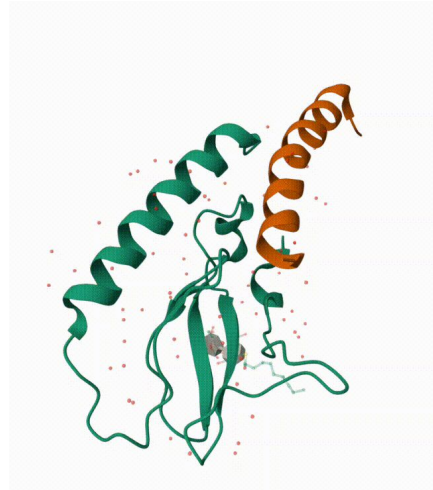
GLP-1 binds to its receptor GLP1R to induce insulin secretion

GLP-1 is secreted from the intestine after nutrient intake. GLP-1 binds to its receptor GLP1R, and the binding event activates intracellular signaling and stimulates glucose-dependent insulin secretion.

GLP1R is a GPCR (G protein-coupled receptor) found on pancreatic β cells and other cell types in the body.

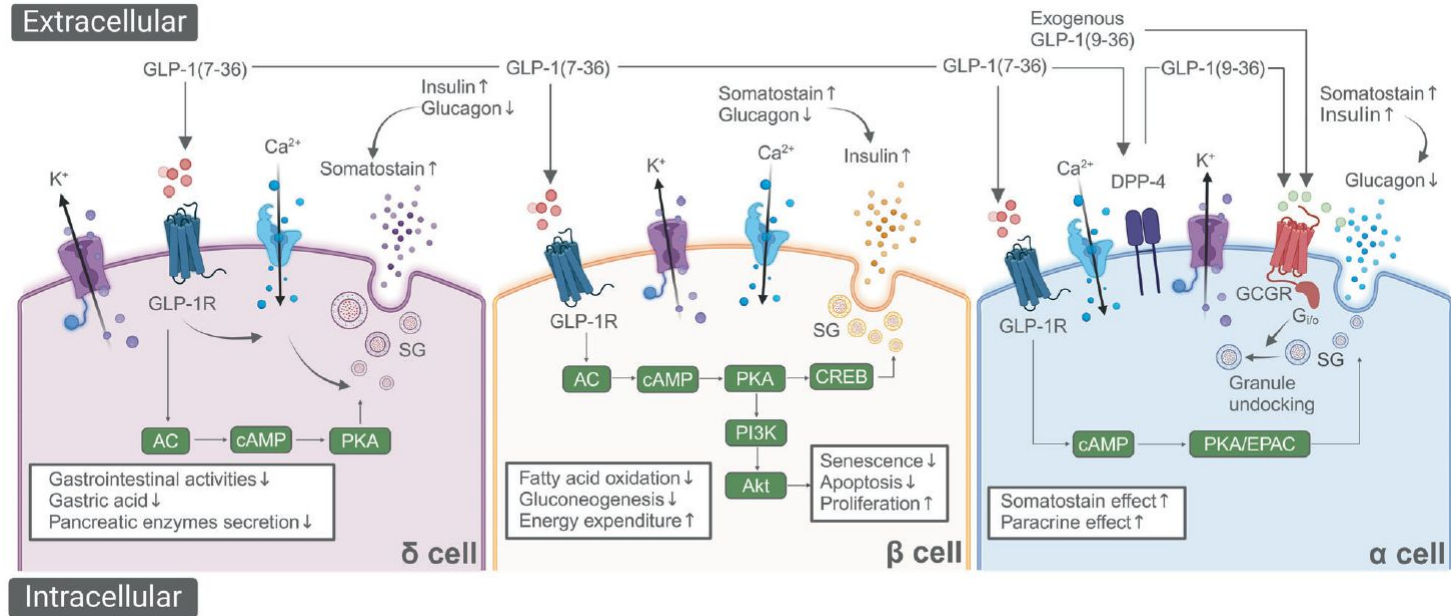
Left: crystal structure of GLP-1 (orange) bound to a fragment of GLP1R (green) (PDB: [3IOL](https://www.rcsb.org/structure/3IOL))

Right: ribbon representation of GLP1R bound to an agonist peptide. The transmembrane domain, or TMD, is coloured cyan. The extracellular domain, or ECD, is coloured brown.



A. Jazayeri et al. (2017), "Crystal structure of the GLP-1 receptor bound to a peptide agonist," <https://doi.org/10.1038/nature22800>, Nature

Mechanisms of blood glucose reduction by GLP-1 in pancreatic α (alpha), β (beta), and δ (delta) cells



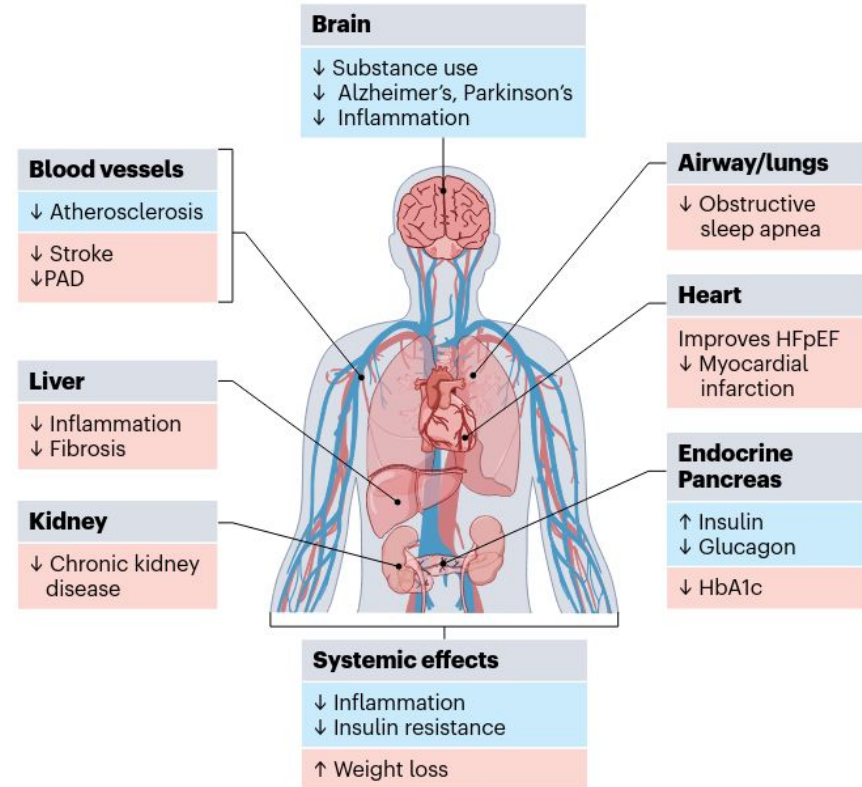
The cartoons give a simplified summary of our current understanding of how GLP-1 reduces blood glucose levels by acting on different pancreatic cell types. Direct and indirect effects of GLP-1 on other cells may employ other mechanisms.

Organ-level effects of GLP-1 medicines: established and potential benefits

The benefits of GLP-1 medicines have been demonstrated in multiple outcome trials for people with

- cardiovascular disease,
- peripheral artery disease (PAD),
- liver disease, kidney disease,
- Heart Failure with preserved Ejection Fraction (HFpEF), and
- obstructive sleep apnea

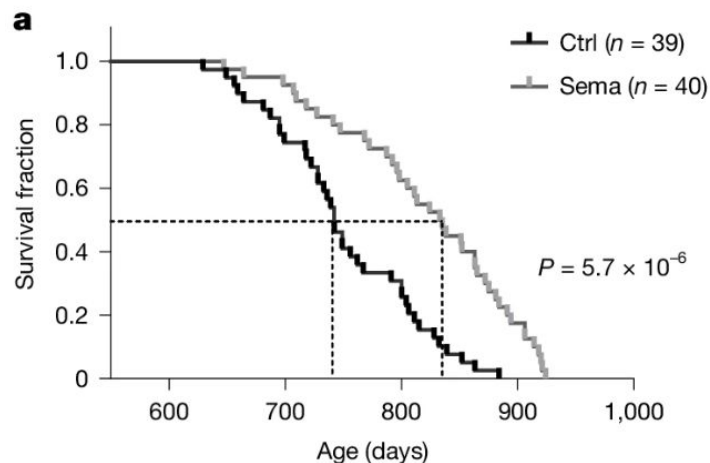
Multiple new GLP-1 medicines are under investigation for a wide range of indications.



Drucker, Daniel J. "The Expanding Landscape of GLP-1 Medicines." *Nature Medicine* 32, no. 1 (2026): 47–57. <https://doi.org/10.1038/s41591-025-04124-5>.

HbA1c: glycated haemoglobin A1c. Oral semaglutide Alzheimer's trials (EVOKE/EVOKE+) did not meet their primary endpoints (reported by the end of 2025).

Late-life semaglutide treatment slows ageing and extends lifespan in female mice



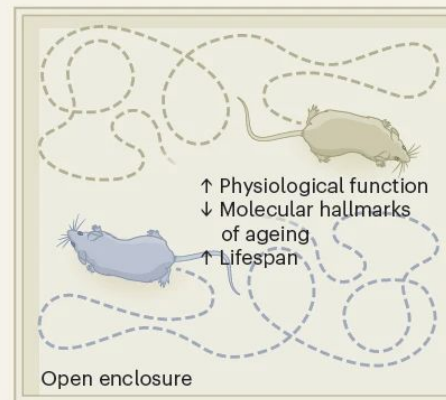
Feng *et al.* put 20-month-old female mice from an inbred laboratory strain on a calorie-restricted diet or treated them with semaglutide. The results suggest that GLP-1 receptor activation and calorie restriction slow ageing through overlapping but distinct mechanisms.

Y. Feng *et al.* (2026), "Late-life semaglutide treatment slows ageing and extends lifespan in female mice," <https://doi.org/10.1038/s41586-026-10940-7>, Nature; News and Views: [Weight-loss drug slows ageing in female mice](#) by Maria Emilia Fernandez & Rafael de Cabo

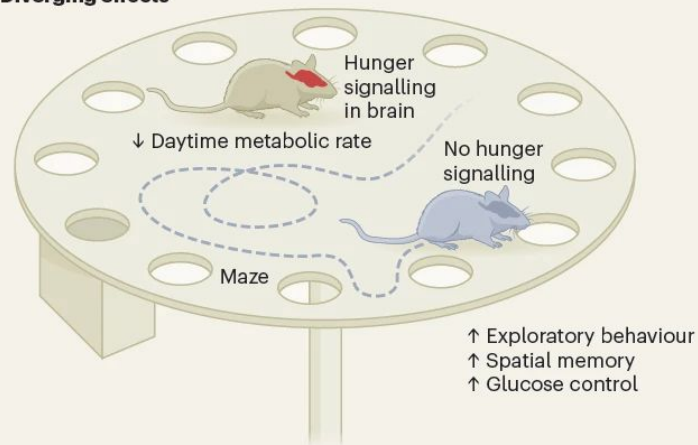
a Condition



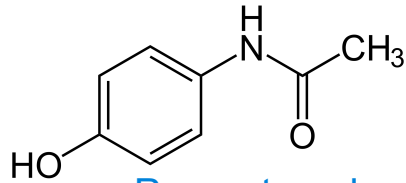
b Converging effects



c Diverging effects

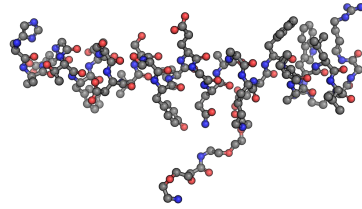


A zoo of modalities



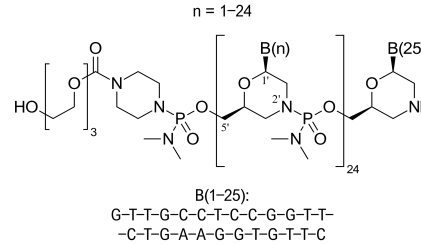
Paracetamol
Molecular weight
(MW): 151 Da

Small molecule



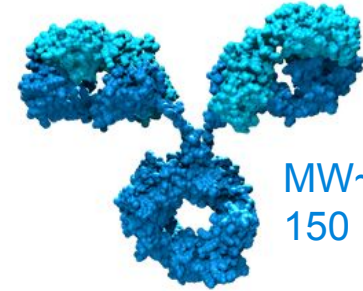
Semaglutide
MW~4 kDa

Peptide



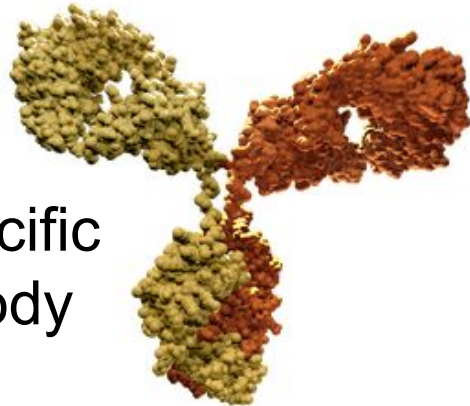
Golodirsen,
MW~5-30 kDa

Oligonucleotides

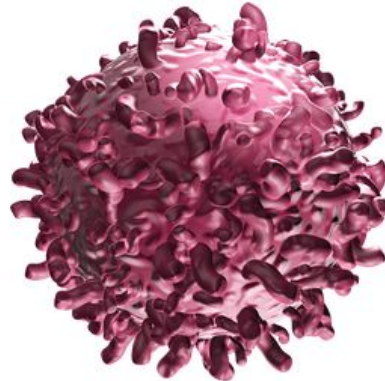


MW~
150 kDa

Monoclonal
antibody



Bispecific
antibody



Chimeric
Antigen
Receptor
(CAR) T
cells



mRNA vaccines

Novel drugs approved by the FDA's Center for Drug Evaluation and Research (CDER) in 2025

- Small molecules usually have a molecular weight (MW) less than 1000 Daltons.
- Oligonucleotides: MW between 5 and 30 kDa (5000-30000 Da), negatively charged
 - siRNA: small interfering RNA
- Proteins: MW ~150 kDa
 - mAb: monoclonal antibody
 - Bispecific: antibodies that bind simultaneously to two antigens or two epitopes of the same antigen.

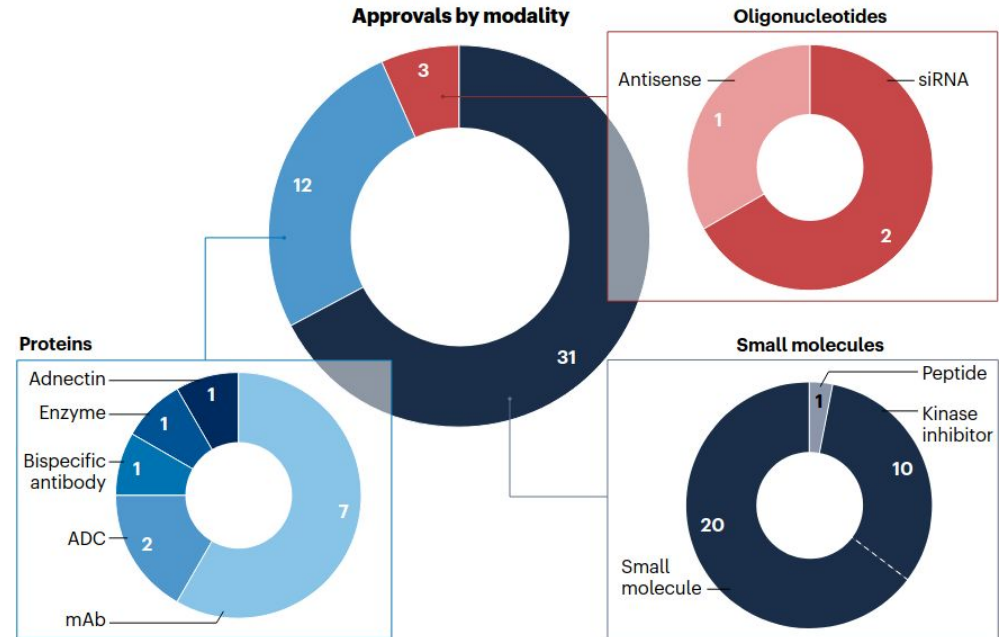
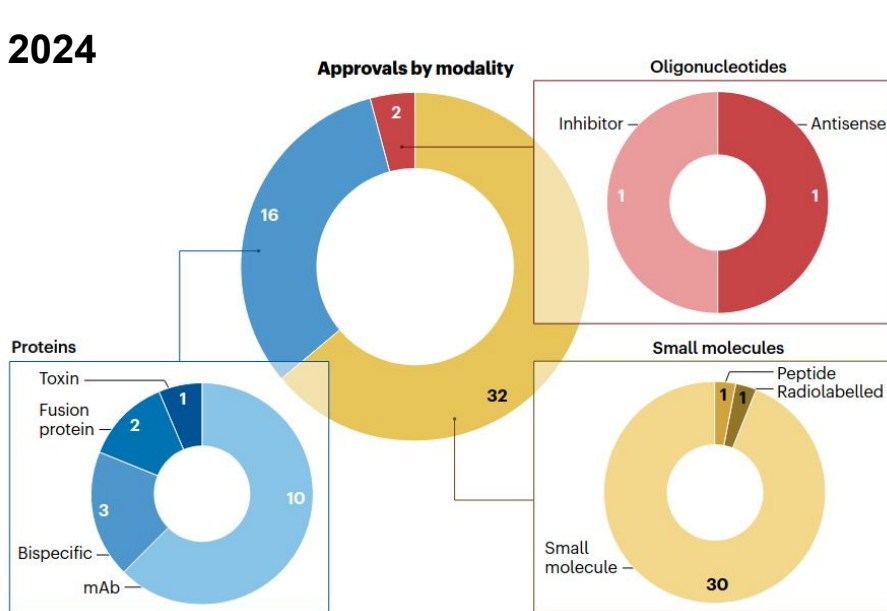


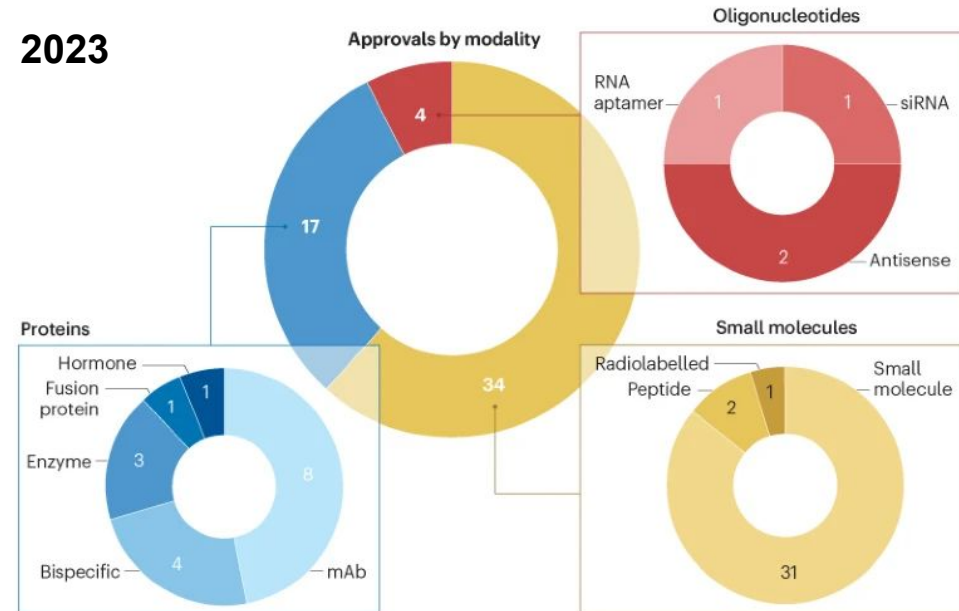
Fig. 3 | CDER approvals by modality. Small molecules, including peptides of up to 40 amino acids in length, and oligonucleotides are approved as new molecular entities. Protein-based candidates are approved through biologics license applications. ADC, antibody–drug conjugate; mAb, monoclonal antibody; siRNA, small interfering RNA. Source: *Nature Reviews Drug Discovery*.

Relative contributions of modalities have remained constant in the past three years

2024



2023



Approval data since 1993

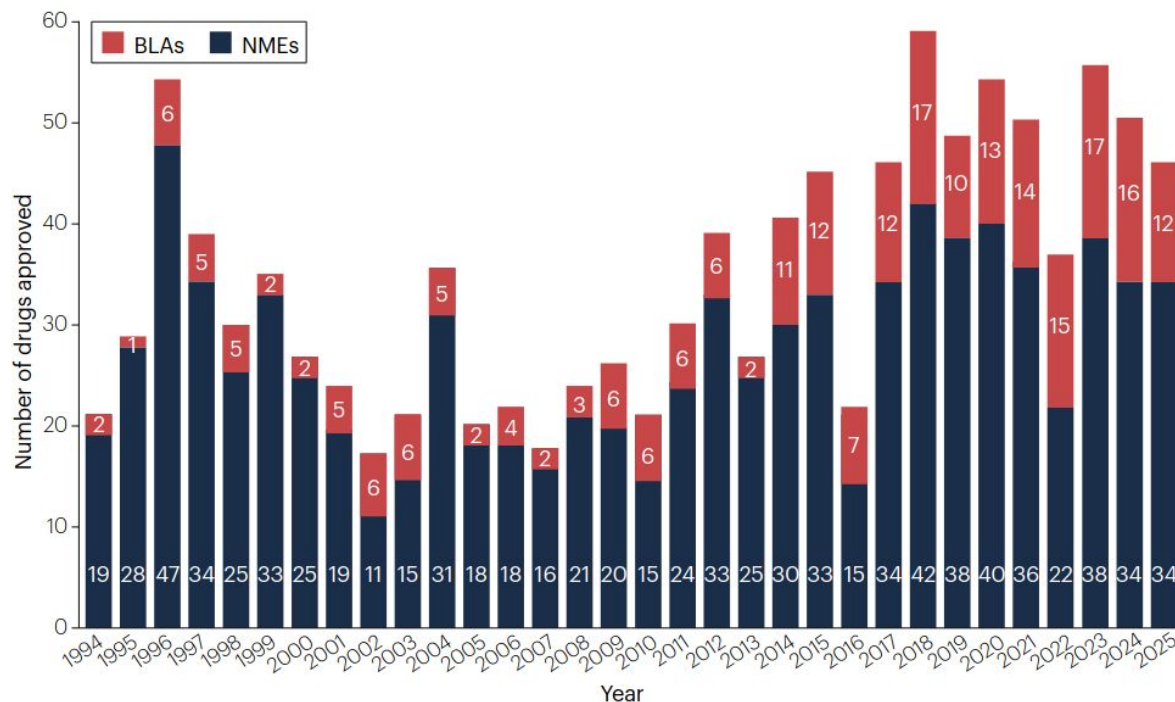


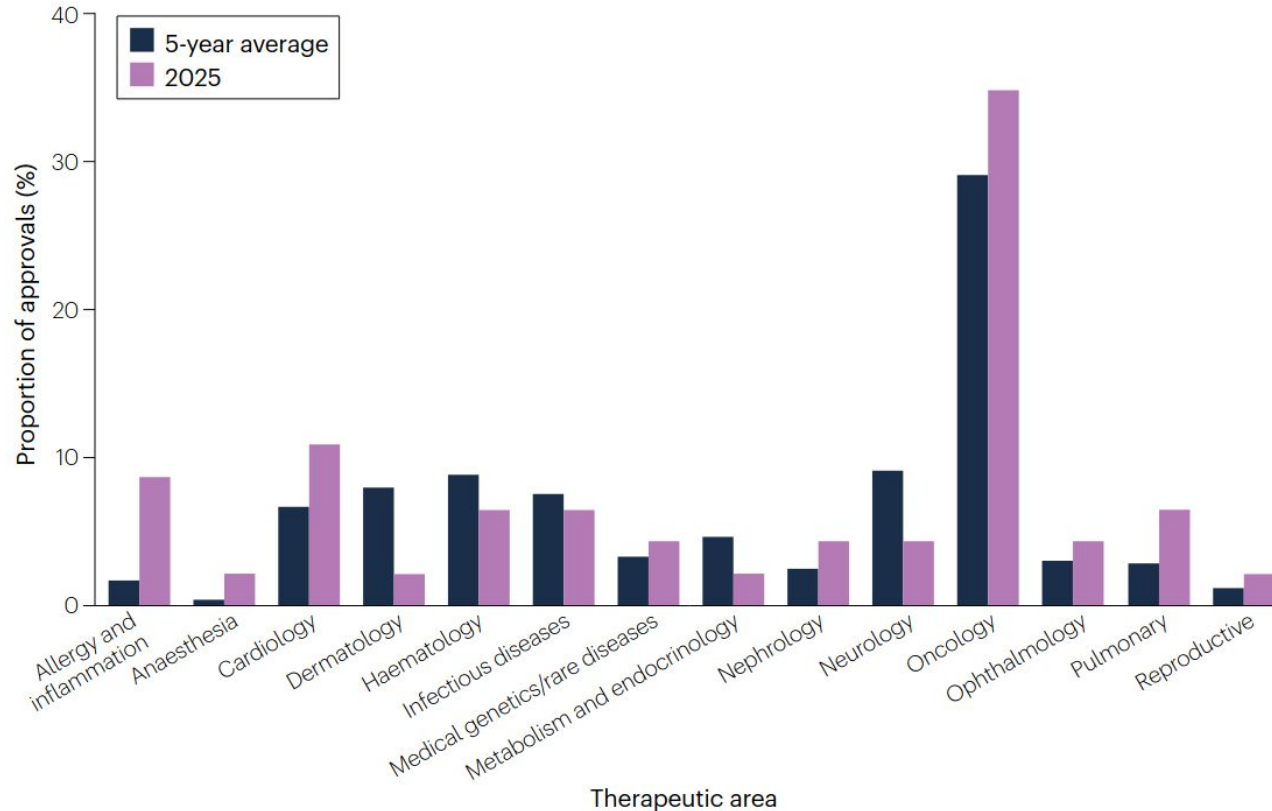
Fig. 1 | Novel FDA approvals since 1993. Annual numbers of new molecular entities (NMEs) and biologics license applications (BLAs) approved by the FDA's CDER. See Table 1 for new approvals in 2025. Products approved by CBER, including vaccines and gene therapies, are not included in this drug count (Table 2). Source: FDA.

Approvals by FDA's Center for Drug Evaluation and Research (CDER):

- BLA: Biologics License Applications (mainly antibodies)
- NME: New Molecular Entities (small molecules, oligonucleotides).

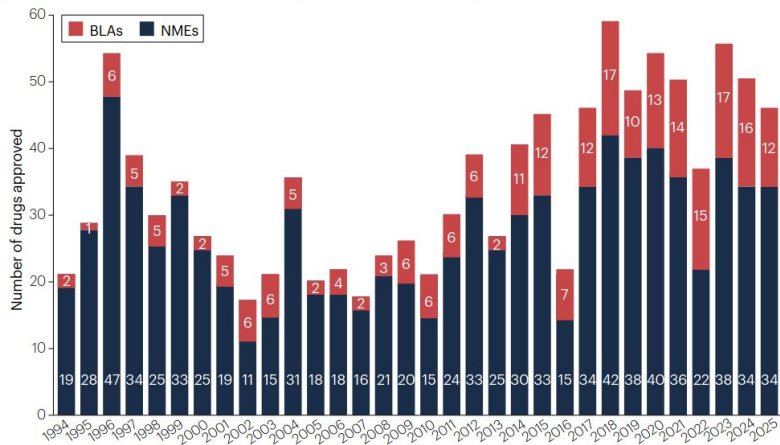
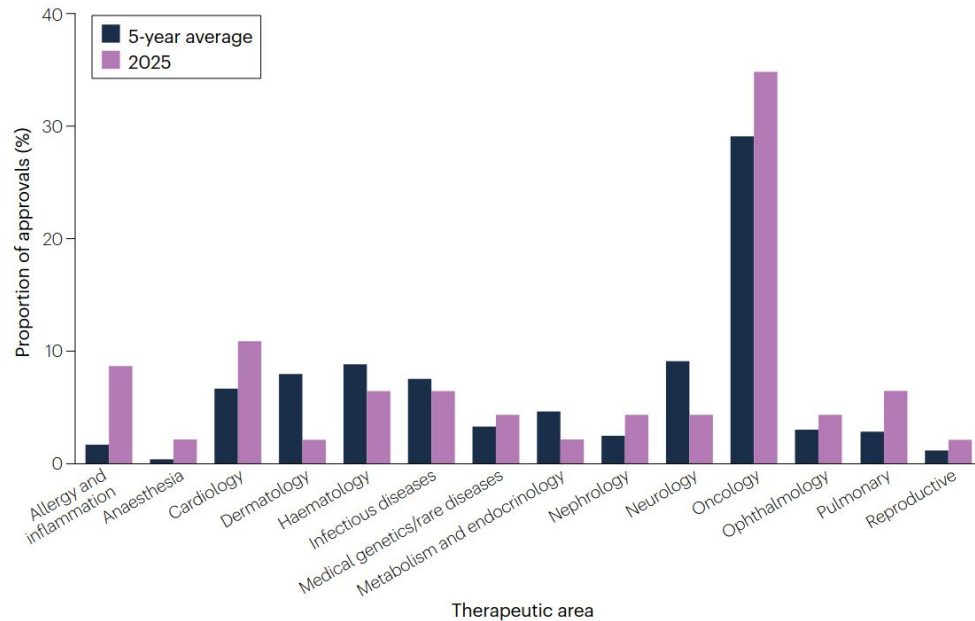
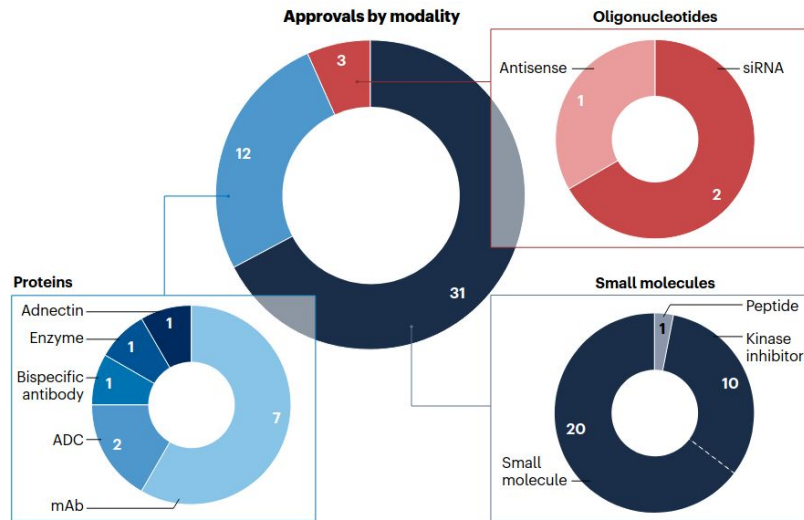
Vaccines and gene therapies are excluded.

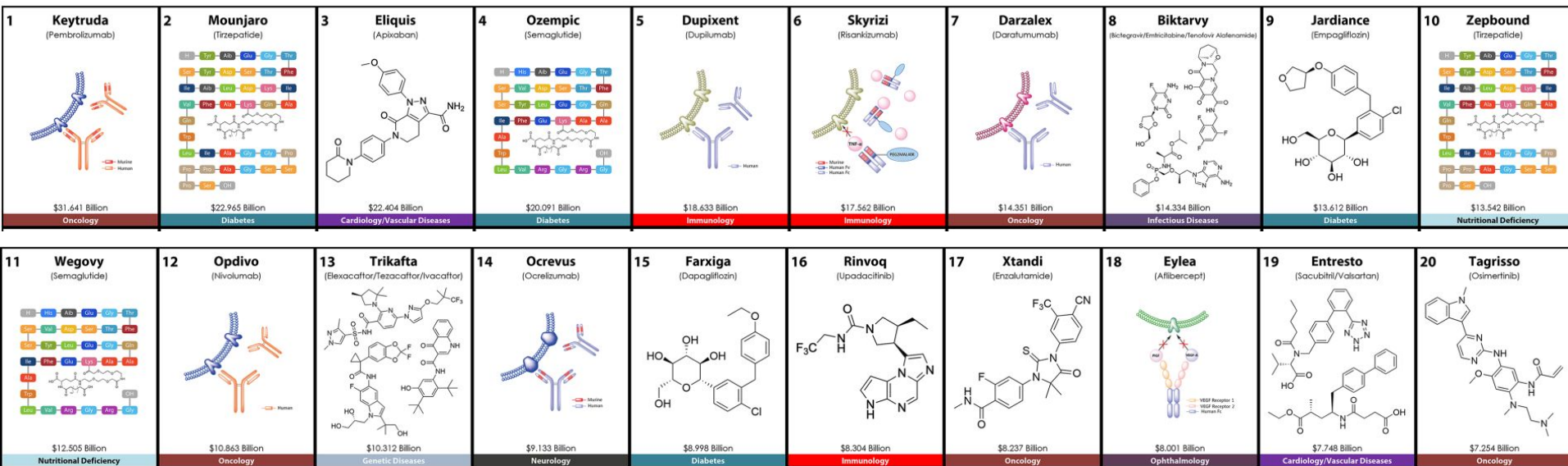
New drug approvals classified by therapeutic areas



- **Cardiology:** heart
- **Dermatology:** skin
- **Hepatology:** liver
- **Haematology:** blood
- **Infectious diseases:** infections due to viruses, bacteria, fungi, etc.
- **Medical genetics/rare diseases:** genetic disorders
- **Metabolism & endocrinology:** metabolism and hormones
- **Nephrology:** kidney
- **Neurology:** brain and peripheral nerves
- **Oncology:** cancer
- **Ophthalmology:** eye
- **Pulmonary:** lung
- **Reproductive:** reproduction and sex

What surprise you?





Top 20 pharmaceuticals by sales in 2025

Poster compiled by the Jon Njardarson group at University of Arizona (<https://njardarson.lab.arizona.edu>). Citation: J. Chem. Ed. 2010, 87, 1348.

Questions: (1) How many of them are small molecules, proteins, and oligonucleotides, respectively ? (2) Are there other modalities? (3) What patterns do you observe? (4) Do you have explanations for these patterns?

The social context of drug discovery: a role-playing game

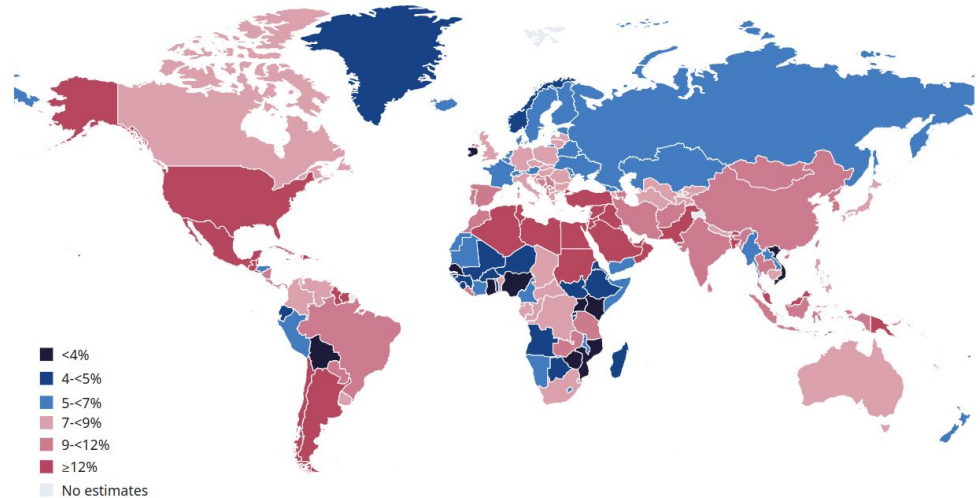
We consider a case study of developing new drugs to treat complications caused by type 2 diabetes, affecting ~10% of adults (20–79 y) worldwide (Switzerland: 4-6%, of whom ~28% are not diagnosed), with the possibility to expand it to the obesity indication.

We divide the classroom into the following personas:

1. Researchers (biology, medicine, AI, etc.)
2. Research funding agency
3. Patients
4. Medical doctors
5. Drug discovery company
6. Insurance company
7. The regulatory agency

Questions: (1) What are your main interests and concerns? (2) With which groups do you wish to collaborate? Why? Rank the partners by priority. (3) What are the best-case and worst-case scenarios for you?

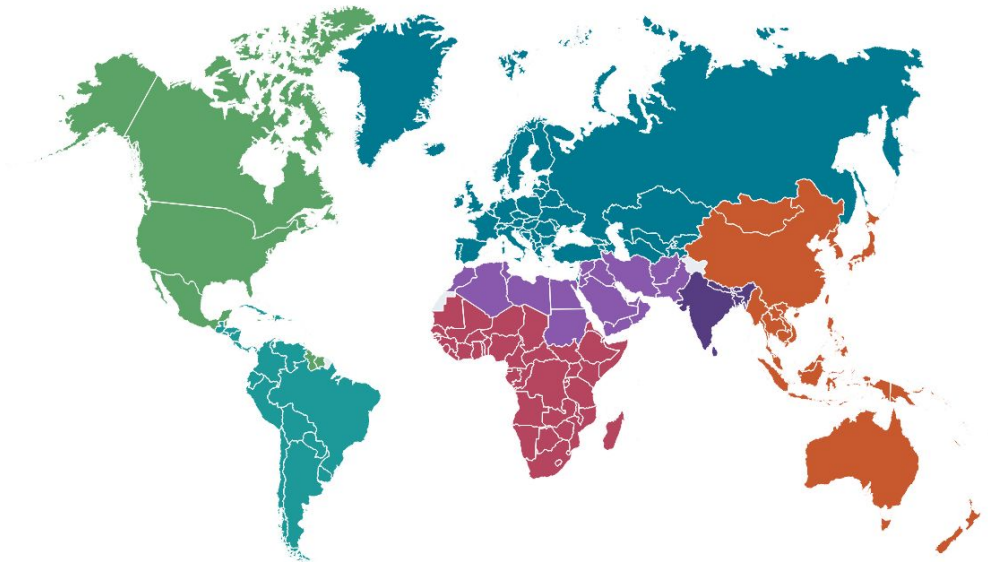
Map 3.2 Estimated age-standardised country prevalence of diabetes in adults (20–79 years), 2024.



[IDF Diabetes Atlas 2025](https://diabetesatlas.org/data-by-location/country/switzerland/) (11th version), The International Diabetes Federation (IDF). Swiss data is acquired from <https://diabetesatlas.org/data-by-location/country/switzerland/>

Statistics and forecast

Map 1 Number of people with diabetes worldwide and per IDF Region, in 2024–2050 (20–79 years)



World		
2050	852.5 Million	↑ 45% increase
2024	588.7 Million	

Europe (EUR)		
2050	72.4 Million	↑ 10% increase
2024	65.6 Million	

North America and Caribbean (NAC)		
2050	68.1 Million	↑ 21% increase
2024	56.2 Million	

South-East Asia (SEA)		
2050	184.5 Million	↑ 73% increase
2024	106.9 Million	

Africa (AFR)		
2050	59.5 Million	↑ 142% increase
2024	24.6 Million	

Middle-East and North Africa (MENA)		
2050	162.6 Million	↑ 92% increase
2024	84.7 Million	

South and Central America (SACA)		
2050	51.5 Million	↑ 45% increase
2024	35.4 Million	

Western Pacific (WP)		
2050	253.8 Million	↑ 18% increase
2024	215.4 Million	

[IDF Diabetes Atlas 2025](#) (11th version)